

A novel agricultural drought index based on multi-source remote sensing data and interpretable machine learning

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ABSTRACT

Drought is a frequent, destructive, and complex natural hazard, and seriously threatens eco-environment, socio-economy, and the health of human. Previous studies suggested that integrated multi-source remote sensing drought indices have the potential to comprehensively monitor drought conditions, however most existing integrated drought indices still have several limitations. Here, we used solar-induced chlorophyll fluorescence, water balance, soil moisture, and land surface temperature to develop a new integrated remote sensing drought index, namely interpretable machine learning drought index (IMLDI), based on the Bayesian optimized tree-based Light Gradient Boosting Machine and SHapley Additive exPlanations. The different land cover types were further considered, and the categories of drought severity were objectively determined by the iterative optimized method. The drought monitoring performance of IMLDI was validated in the eastern parts of China, and three integrated drought indices composed by PCA, multiple linear regression, and gradient boosting method were also included for comparison. The results show that IMLDI has a higher spatial and temporal consistency with standardized precipitation evapotranspiration index, can better reflect the real-world observed drought-affected cropland areas and gross primary production, and can also well describe the evolutions of 2009/2010 and 2019 drought events in the eastern parts of China, indicating higher drought monitoring performance of IMLDI. Besides, IMLDI-based agricultural drought risk analysis shows that the Huang-Hai Region and Yunnan, Guizhou, and Guangxi Provinces have a high risk to suffer from severe agricultural droughts. Overall, IMLDI has a great potential to use as a new integrated remote sensing drought index for agricultural drought monitoring.

1. Introduction

Drought is a frequent, destructive, long-lasting, and complex extreme weather event (Liu et al., 2023a; Pradhan et al., 2017; Zhang et al., 2017; Zhang and Yuan, 2020). Drought is generally classified into meteorological drought, agricultural drought, hydrological drought, and socioeconomic drought (Dracup et al., 1980; Wilhite and Glantz, 1985). Meteorological drought is a specified period with an imbalance between precipitation and evaporation. Agricultural drought occurs when soil

moisture deficits below the levels necessary for sustaining plant growth, and usually leads to substantial crop yield reductions and severe economic repercussions (Jiao et al., 2019; Mishra and Singh, 2010). Hydrological drought occurs when rivers and other water bodies fall below their normal levels or the water level of underground aquifers. Socio-economic drought refers to an insufficient water supply to meet the economic demands. Under background of global warming, drought became much more frequent and intensified in most parts of the world during recent decades, which poses a serious threat to the health of

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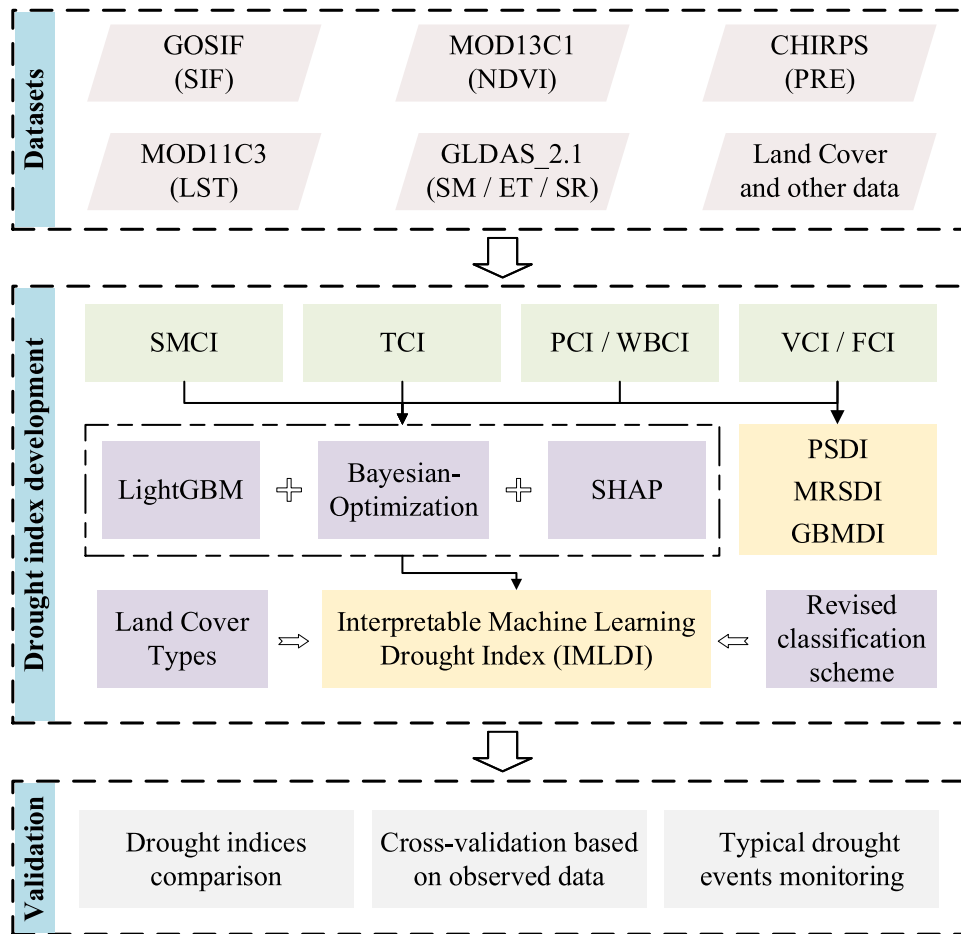


Fig. 1. The technical flowchart.

human and eco-environment, and causes direct economic losses of billions of dollars every year (AghaKouchak et al., 2015; Gao et al., 2019; Naumann et al., 2021; Verhoeven et al., 2022). For example, according to the “China Flood and Drought Disaster Bulletin”, China experienced a severe drought disaster in 2019, which covers 18.23 million ha agricultural area and causes a shortage of drinking water for 6.92 million people and 3.68 million livestock. Therefore, it is important to develop effective drought indices to accurately monitor drought conditions and understand regional drought risks (Rhee and Im, 2017; Samantaray et al., 2022).

Numerous drought indices have been developed in recent years, such as Palmer Drought Severity Index (PDSI) (Palmer, 1965), Standardized Precipitation Index (SPI) (McKee et al., 1993), and Standardized Precipitation Evapotranspiration Index (SPEI) (Vicente-Serrano et al., 2010). As an improved version of SPI, SPEI is defined by the difference between evapotranspiration and precipitation (Zhang and Jia, 2013; Zhang et al., 2018b). SPEI is a multi-scalar drought index for quantifying the onset, duration, and magnitude of drought, and is better than PDSI in monitoring short-term droughts (Hao et al., 2015; Hao and Singh, 2015). These traditional drought indices are generally calculated based on ground-based observed meteorological variables. Ground-based observations can provide reliable observed data, but cannot capture spatial and temporal distributions of drought due to the sparse and uneven distributions (Beck et al., 2017; Jiao et al., 2019; Sun et al., 2017; West et al., 2019).

Remote sensing can obtain spatiotemporal continuous data of long time series in near real time and provides a new way to monitor drought. During recent years, many remote sensing drought indices, such as Normalized Difference Vegetation Index (NDVI) (Rouse et al., 1974),

Enhanced Vegetation Index (EVI) (Huete et al., 2002), Vegetation Condition Index (VCI) (Kogan, 1995a), Temperature Condition Index (TCI) (Kogan, 1995b), Normalized Difference Water Index (NDWI) (Gao, 1996), Precipitation Condition Index (PCI) (Du et al., 2013), Soil Moisture Condition Index (SMCI) (Zhang and Jia, 2013), and Vegetation Drought Index (VDI) (Sun et al., 2013), have been developed to monitor regional or global drought conditions (Lu et al., 2016; Rhee et al., 2010; Wang et al., 2012; Wu et al., 2013). However, these single hydrometeorological variable-based drought indices cannot fully and reliably capture drought conditions because of the complexity processes and diverse impacts of drought (Guo et al., 2019; Tian et al., 2018).

As more drought-related variables are successfully retrieved from remote sensing measurements, many integrated multi-sources remote sensing drought indices have also been developed to better comprehensively assess and monitor drought conditions (Li et al., 2020a, 2020b). The common used and representative integrated remote sensing drought indices are the Vegetation Health Index (VHI) (Kogan et al., 2004), scaled drought condition index (SDCI) (Rhee et al., 2010), Microwave Integrated Drought Index (MIDI) (Zhang and Jia, 2013), Synthesized Drought Index (SDI) (Du et al., 2013), Optimized Vegetation Drought Index (OVDI) (Hao and Singh, 2015), Process-based Accumulated Drought Index (PADI) (Zhang Xiang et al., 2017), Integrated Remote Sensing Drought Monitoring Index (IRSDI) (Sun et al., 2017), Geographically Independent Integrated Drought Index (GIIDI) (Jiao et al., 2019), Temperature Vegetation Precipitation Dryness Index (TVPDI) (Wei et al., 2020), Temperature Fluorescence Drought Index (TFDI) (Zhang et al., 2020), Fluorescence Healthy Vegetation Index (SHI) (Liu et al., 2021a), Bivariate Soil Moisture and Evapotranspiration Index (BSMEI) (Wu et al., 2021), drought index integrated by gradient

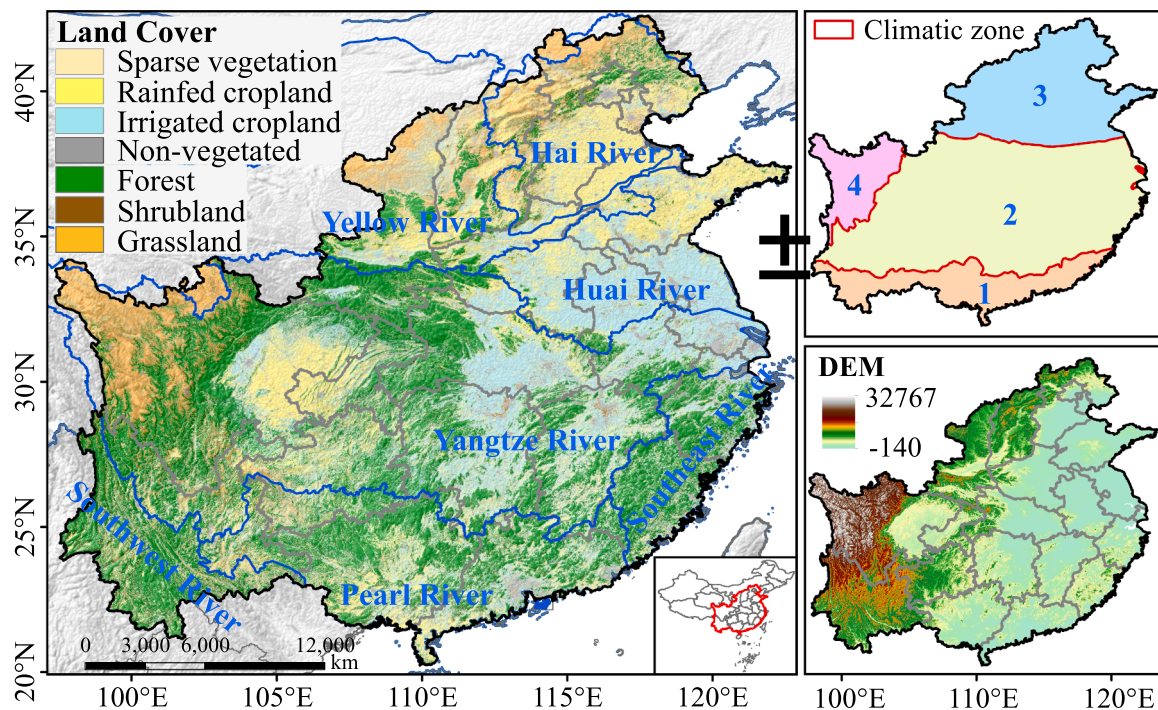


Fig. 2. Spatial distribution of land cover types in the eastern parts of China.

boosting method (GBMDI) (Zhang et al., 2022), Temperature-vegetation-soil-moisture-precipitation-dryness Index (TVMPDI) (Wang et al., 2022), temperature-sun-induced chlorophyll fluorescence-water balance dryness index (TSWDI) (Liu et al., 2023b), Geographically Integrated Dryness Index (GIDI) (Wei et al., 2023), and Combined Drought Index (CDI) (Başakın et al., 2024). However, these integrated drought indices still have several limitations when applied to drought monitoring.

These integrated remote sensing drought indices were generally developed based on NDVI, precipitation, and other variables. One limitation is that NDVI is easy to be affected by the soil background and saturated especially in high vegetation coverage areas, and precipitation cannot fully assess the wet and dry conditions of the climate (Birky, 2001; Hmimina et al., 2013; Liu et al., 2021a, 2021b; Sui et al., 2013). The solar-induced chlorophyll fluorescence (SIF) is directly related to the internal photosynthesis mechanism of plants and is less affected by soil background and clouds, and thus can directly reflect the dynamics of actual plant photochemical production (Cui et al., 2017; Guanter et al., 2014; Lee et al., 2013). Many previous studies suggested that SIF is much more sensitive to drought stress than NDVI (Daumard et al., 2010; Liu and Yue, 2018; Yoshida et al., 2015; Zhang et al., 2019a, 2019b). The water balance (WB) is constructed from surface runoff, precipitation, and evapotranspiration, which provides a more accurate reflection of climatic wet and dry conditions than precipitation (Liu et al., 2023b; Sheffield et al., 2009).

Another issue is that many previous drought indices were developed without considering different land cover types (Malczewski and Liu, 2014; Tang et al., 2018). However, drought conditions are significantly influenced by land cover types because different vegetation species have different physiological mechanisms and adaptation processes to drought stresses (Abbas et al., 2014; Hua et al., 2017; Lamchin et al., 2018; Li et al., 2018a, 2018b). Additionally, drought severity categories of most previous drought indices were arbitrarily classified, which may overestimation or underestimation of the actual drought conditions (Guo et al., 2019; Hao et al., 2015).

There are many different methods of generating integrated remote sensing drought indices, such as the linear combination, principal

components analysis (PCA), joint distribution methods, and machine learning models (Xu et al., 2023). Among these methods, the linear combination method is the simplest and most easily applied method, but determining the weights of different variables objectively is very difficult (Cai et al., 2017; Hao and Singh, 2015). Machine learning models, such as tree-based Light Gradient Boosting Machine (LGBM), have been proved to be powerful and flexible methods in many linear and nonlinear regression applications, and provide a new method to develop integrated drought indices (Feng et al., 2019; Park et al., 2016; Rhee and Im, 2017). However, machine learning models have a complex structure and many parameters, which leads to high difficulty in parameter tuning and overfitting (Alibrahim and Ludwig, 2021; Feurer and Hutter, 2019). Moreover, machine learning models are regarded as black-box models, which makes it difficult to understand why the models make a particular regression or prediction, thus limiting the model trustworthiness and scope of application (Li, 2022; Van den Broeck et al., 2022).

To address these limitations, this study aims to propose a new interpretable machine learning drought index (IMLDI) based on the following considerations: (i) SIF, WB, soil moisture (SM), and land surface temperature (LST) variables are used, (ii) drought index is developed for different land cover types, (iii) drought index is composited by the Bayesian optimized LGBM model and interpreted with SHapley Additive explanations (SHAP), and (iv) drought severity categories are objectively classified by iterative optimization method. The drought monitoring performance of IMLDI was evaluated in the eastern parts of China, where observed and statistical data can serve as a reference for validation.

2. Material and methods

The technical flowchart is shown in Fig. 1, including data collection, drought index development, and verification.

2.1. Study area

The eastern parts of China (Fig. 2) cover 37.12 % of China's land area (22° - 42°N, 97° - 135°E), and the main land cover types are forest,

scrubland, grassland, sparse vegetation, rainfed cropland, irrigated cropland, and non-vegetated areas. It locates in East Asian Monsoon region, where climate is characterized by pronounced shift in wet and dry seasons. The study area contains the marginal tropical humid (sub-region 1), northern subtropical humid (sub-region 2), warm temperate semi-humid (sub-region 3), and plateau temperate semi-arid (sub-region 4) climatic zones. The eastern parts of China are low in east and high in west, and mainly include the Yangtze River, Pearl River, Yellow River, Huai River, Hai River, Southwest River, and Southeast River basins. In recent decades, the eastern parts of China suffered from extreme drought events frequently (Chen et al., 2023; Shi et al., 2020; Yu and Zhai, 2020).

2.2. Data

2.2.1. Observed meteorological data

The CN05.1 gridded monthly precipitation and temperature datasets with a spatial resolution of $0.25^\circ \times 0.25^\circ$ during 2001–2020 were used, which is interpolated from in situ observed precipitation and temperature at approximately 2400 meteorological stations in China using the anomaly approach method. Strict quality control has also been implemented before these datasets release.

2.2.2. Remote sensing data

In this work, the monthly global OCO-2-based Sun-induced Chlorophyll Fluorescence (GOSIF) (Li and Xiao, 2019a), GOSIF gross primary production (GPP), Moderate Resolution Imaging Spectroradiometer products (MODIS), and the Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) datasets derived from remote sensing measurements, and the monthly Global Land Data Assimilation System Version 2 (GLDAS-2.1) Noah model simulation products were used. Table S1 provides detailed information about these remote sensing and assimilation datasets. All datasets are collected over the period from 2001 to 2020.

GOSIF was developed based on discrete OCO-2 SIF soundings, MODIS remote sensing data, and MERRA-2 meteorological reanalysis data using the Cubist model (Li and Xiao, 2019a), which has good spatial resolution, continuous global coverage, and long records (Qiu et al., 2022; Shekhar et al., 2022). GOSIF GPP dataset was developed by using the linear relationships between SIF and GPP (Li and Xiao, 2019b). MODIS NDVI (MOD13C1) and LST (MOD11A2) with a spatial resolution of 0.05° were obtained from National Aeronautics and Space Administration's Earth Observing System Data and Information System. CHIRPS is a quasi-global (50°S – 50°N) long-term precipitation dataset, which is developed by the United States Geological Survey and Climate Hazards Group (Funk et al., 2015a, 2015b, 2014). The GLDAS-2.1 products were derived from station-based and satellite-based observations using data assimilation algorithms. Previous studies suggested that spatiotemporal continuous SM, evapotranspiration (ET), and runoff (SR) from GLDAS can well capture the spatial and temporal variations of corresponding observed hydrometeorological variable (Jasinski et al., 2019; Kumar et al., 2019; Wei et al., 2021). Therefore, monthly 0–10 cm depth SM, ET, and SR from GLDAS-V2.1 Noah Model L4 were used.

2.2.3. The statistical drought-affected cropland areas and Land cover data

The statistical drought-affected cropland areas data during 2001–2020 was sourced from the crop databank of the Chinese Crop Farming Information center at <https://data.stats.gov.cn/>. These statistical records were used to validate the drought monitoring performance of the newly developed drought index.

The land cover types data was obtained from the European Space Agency for the Climate Change Initiative (ESA CCI) (Li et al., 2018c). This land cover types data consists of 37 land cover types with a resolution of 300 m, which has a higher overall accuracy in China than many other land cover types products (Gessner et al., 2015; Hartley et al., 2017; Li et al., 2018c; Yang et al., 2017). The original ESA CCI land cover types were re-grouped into forest, shrubland, grassland, sparse

Table 1

Descriptions of scaled remote sensing drought indices.

Drought indices	Formula	Ref.
VCI	$\frac{NDVI_i - NDVI_{\min}}{NDVI_{\max} - NDVI_{\min}}$	(Kogan, 1995a)
FCI	$\frac{SIF_i - SIF_{\min}}{SIF_{\max} - SIF_{\min}}$	(Zhang et al., 2021)
PCI	$\frac{P_i - P_{\min}}{P_{\max} - P_{\min}}$	(Zhang and Jia, 2013)
WB	$WB = P - Q - E$	(Sheffield et al., 2009)
WB CI	$\frac{WB_i - WB_{\min}}{WB_{\max} - WB_{\min}}$	(Liu et al., 2023b)
SMCI	$\frac{SM_i - SM_{\min}}{SM_{\max} - SM_{\min}}$	(Zhang and Jia, 2013)
TCI	$\frac{LST_{\max} - LST_i}{LST_{\max} - LST_{\min}}$	(Kogan, 1995b)

vegetation, rainfed cropland, and irrigated cropland in the eastern parts of China. Note that all grid datasets used were resampled to a 0.5° grid.

2.3. Methods

VCI, SIF condition index (FCI), PCI, water balance condition index (WB CI), SMCI, and TCI were taken as input variables to develop multiple remote sensing drought indices. Integrated remote sensing drought indices composed by PCA (PSDI), multiple linear regression (MRSDI), and gradient boosting method (GBMDI) were used to validate the drought monitoring performance of IMLDI. Note that PSDI, MRSDI, GBMDI, and IMLDI were developed for each month in a year.

2.3.1. Standardized precipitation evapotranspiration index

SPEI is calculated from the difference between precipitation and temperature, which is widely used to assess drought conditions (Vicente-Serrano et al., 2010). It is often used to validate other drought indices (Park et al., 2017; Um et al., 2018; Zhang et al., 2013). SPEI preserves multiple time scales to indicate different types of droughts, such as meteorological and agricultural droughts. Previous studies suggested that the 3-month SPEI could better monitor the evolutions of agricultural drought in China (Mishra and Singh, 2010; Zambrano et al., 2016, 2017), and thus 3-month SPEI was calculated from observed meteorological data in this work.

2.3.2. Scaled remote sensing drought indices

NDVI, SIF, precipitation (P), WB, SM, and LST were used to calculate scaled VCI, FCI, PCI, WB CI, SMCI, and TCI, respectively. The calculation formulas are shown in Table 1, wherein $*_i$, $*_{\max}$, and $*_{\min}$ are the pixel values, maximum values, and minimum values of variable $*$, respectively, in the same month during 2001–2020. The values of the scaled remote sensing drought indices range from 0 to 1. A scale value of 0 represents the driest conditions, and 1 represents the wettest conditions (Wei et al., 2020).

2.3.3. Integrated remote sensing drought indices for comparison

PCA is a linear transformation reducing redundancy by translating and/or rotating the axes of the original feature space, so that drought information can be represented without correlation in a new component space (Du et al., 2013; Lasaponara, 2006; Zhang et al., 2022). PCA was used to composite monthly VCI, PCI, SMCI, and TCI remote sensing variables, and the first principal component of the PCA output for each month in a year is taken as PSDI (Liu et al., 2019).

MRSDI was developed by combining independent VCI, PCI, and TCI remote sensing variables, and the coefficients for each month in a year (Table S2) were determined by multiple linear regression model (Grégoire, 2014; Noi et al., 2017). In this case, SMCI is the major mechanism behind agricultural drought, which was used as the dependent variable (Sun et al., 2017; Zhang et al., 2018a, 2022).

Gradient boosting machine (GBM) can significantly improve

Table 2
Drought severity categories of SPEI, PSDI, MRSDI, GBMDI, IMLDI.

Drought categories	SPEI	PSDI	MRSDI	GBMDI	IMLDI
Extreme drought	≤ -2	≤ 0.12	≤ 0.4	≤ 0.10	≤ 0.14
Severe drought	$-2 \sim -1.50$	$0.12 \sim 0.24$	$0.40 \sim 0.46$	$0.10 \sim 0.20$	$0.14 \sim 0.24$
Moderate drought	$-1.50 \sim -1$	$0.24 \sim 0.36$	$0.46 \sim 0.52$	$0.20 \sim 0.30$	$0.24 \sim 0.34$
Mild drought	$-1 \sim 0$	$0.36 \sim 0.48$	$0.52 \sim 0.58$	$0.30 \sim 0.40$	$0.34 \sim 0.52$
No drought	≥ 0	≥ 0.48	≥ 0.58	≥ 0.40	≥ 0.52

prediction accuracy, optimize computational efficiency, and effectively avoid overfitting (De'Ath, 2007; Zhang and Haghani, 2015). GBMDI is integrated remote sensing drought index based on the GBM (Zhang et al., 2022). Monthly VCI, PCI, SMCI, and TCI as the input variables,

and monthly SPEI was taken as the reference data. The scaled relative importance of these input variables for each month in a year (shown in Table S3) was used as weights to develop GBMDI. Drought severity categories of PSDI, MRSDI, and GBMDI are given in Table 2.

2.3.4. Light gradient boosting machine, Bayesian hyperparameter optimization, and SHapley Additive exPlanation

LGBM is an ensemble learning model based on boosting strategy, and is an efficient implementation of gradient boosting decision tree (Ke et al., 2017). It incrementally improves the predictive accuracy of the entire model by successively adding new decision trees capable of predicting the residuals of previous models and using the negative gradient of the loss function as a guide. Compared with its counterpart GBM and eXtreme Gradient Boosting (XGBoost), LGBM can achieve higher training efficiency and lower memory footprint while maintaining high prediction accuracy (Bentéjac et al., 2021; Chen and Guestrin, 2016; Szczepanek, 2022).

In machine learning models, the number (complexity) of

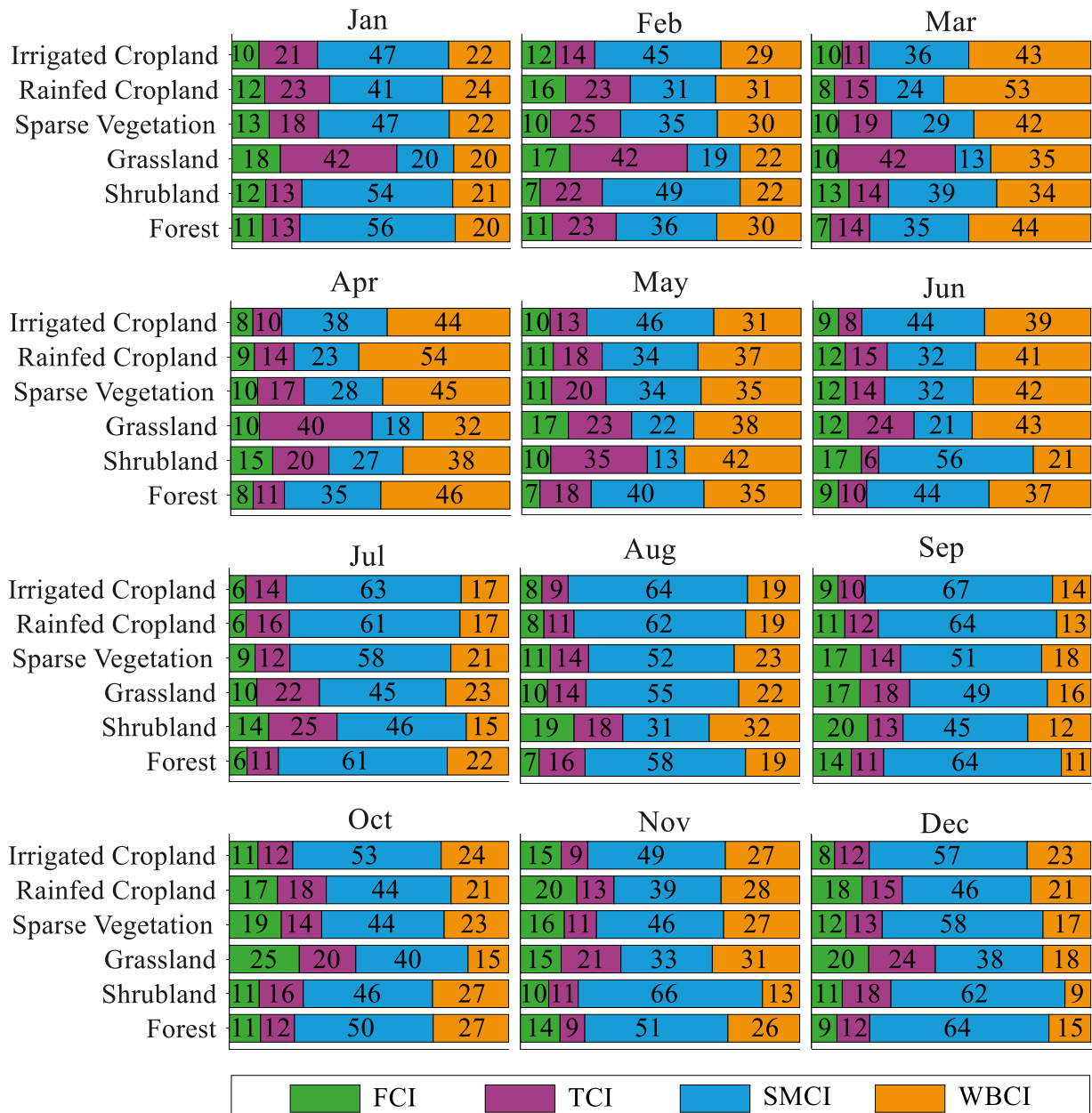


Fig. 3. Relative contributions (%) of FCI, WBCI, SMCI, and TCI to SPEI for each month in six different land cover types.

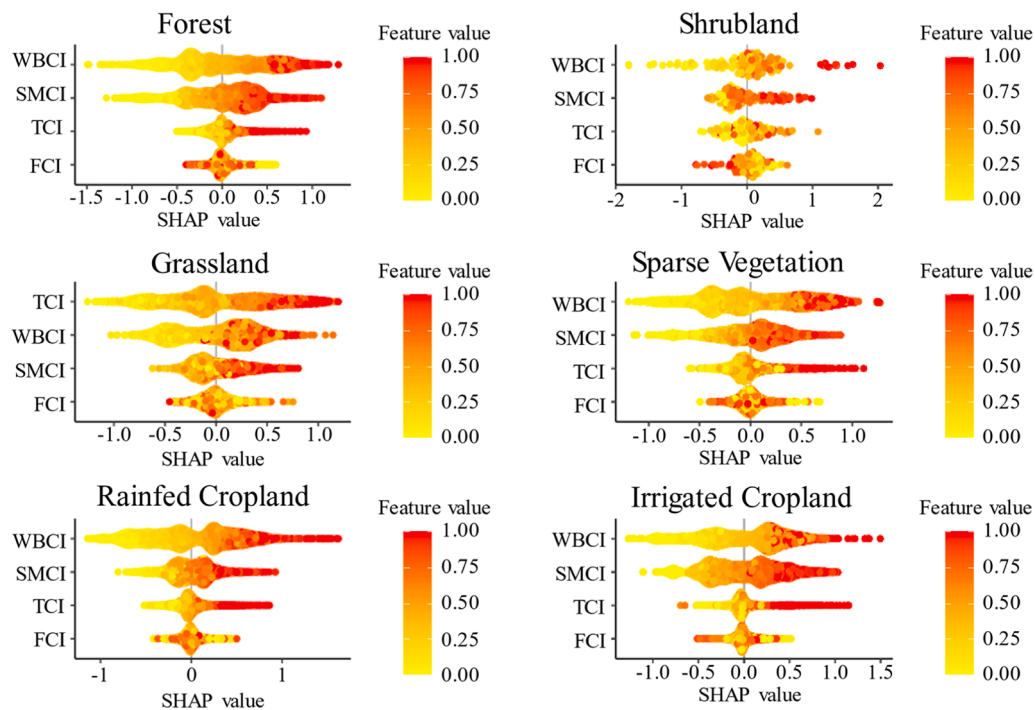


Fig. 4. The SHAP beeswarm plots of the input remote sensing variables contributions to SPEI in different land cover types during April.

hyperparameters is often large (high), and traditional grid search and stochastic search methods are difficult to find the optimal solutions in a reasonable time (Alibrahim and Ludwig, 2021; Feurer and Hutter, 2019). The Bayesian hyperparameter optimization is an efficient, global, and adaptive optimization method, and can find a near-optimal combination of hyperparameters in a smaller number of iterations (Snoek et al., 2012; Wu et al., 2019). Previous studies suggested that the Bayesian optimized LGBM model can form a more powerful machine learning framework (Zhou et al., 2024).

As a machine learning model interpretation tool based on a game theory approach, SHAP provides global and local interpretability for the model by calculating the value of each feature's contribution to the model regression or prediction results (Mokhtari et al., 2019). The SHAP value can be used as a measure of the importance of features, which helps users to understand which features have the greatest impact on the model's regression or prediction results (Yang et al., 2021).

2.3.5. Iterative optimized method

To objectively classify drought severity categories, the iterative optimized method (Um et al., 2018) was adopted to find more meaningful breakpoints of IMLDI by minimizing the difference in the sum of the drought frequency and drought area between SPEI and IMLDI. IMLDI was categorized into four drought categories to better compare with other previous drought indices. For each category, drought frequency error is calculated as the total sum of squares of the relative frequency errors in each pixel, and drought area error is calculated as the total sum of squares of the area errors for each month. The total error is summarized from 8 cases, including the four drought categories for the two types of errors (4 categories $\times$ 2 errors). The optimized thresholds were chosen as the minimum total error by iterating the categorical threshold values with a step of 0.01.

2.3.6. Statistical methods

The Pearson correlation analysis was used to quantify the correlations in validation of IMLDI. When the p-value is less than 0.05 (0.01), the correlation is significant at 95 % (99 %) confident level (Ren et al., 2024).

The theory of runs (Yevjevich, 1967) was used to express the duration, severity, and peak of agricultural drought events in IMLDI-based agricultural drought risk analysis. The threshold value of the IMLDI was set at 0.52 to identify the beginning and end of an agricultural drought event.

3. Results

3.1. Development of interpretable machine learning drought index (IMLDI)

There are five steps to develop IMLDI. (1) FCI, WBCI, SMCI, and TCI were selected as the input remote sensing variables, and SPEI was taken as the reference data. (2) IMLDI was developed for each land cover type. (3) LGBM optimized by Bayesian hyperparameter optimization was used to construct IMLDI, and the importance of input remote sensing variables was interpreted by SHAP. (4) The scaled mean SHAP values of input remote sensing variables were taken as weights to composite IMLDI. (5) Drought severity categories of IMLDI were objectively classified by the iterative optimized method, and are shown in Table 2.

The scaled mean SHAP values (or weights in another word) of input remote sensing variables in the development of IMLDI are given in Fig. 3. For July to February next year, SMCI is a dominant contributor to drought in most land cover types. This may be because the reduction in soil moisture cannot provide enough water for vegetation and finally triggers agricultural drought (Li et al., 2020b; Zhang and Jia, 2013). However, WBCI dominates drought in most land cover types from March to June. The relative contribution of TCI to SPEI exceeds 40 % in grassland from January to April, which is much greater than that of SMCI, WBCI, and FCI. The contributions of FCI to drought are relatively low compared to other input variables, which agrees with the results from Zhang et al. (2022). However, FCI contributes to a relative higher extent in grassland, sparse vegetation, and shrubland compared to other land cover types. Previous studies also found similar results when exploring the correlations of VCI and SPEI in different land cover types (Vicente-Serrano, 2007).

Moreover, April and August were taken as examples to show the

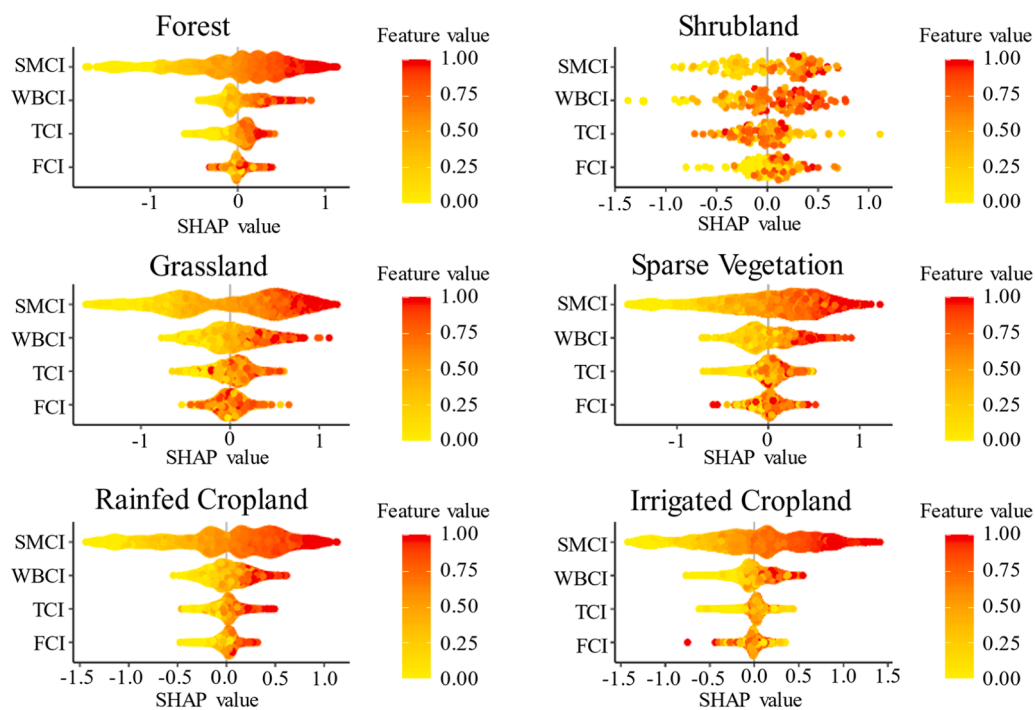


Fig. 5. The SHAP beeswarm plots of the input remote sensing variables contributions to SPEI in different land cover types during August.

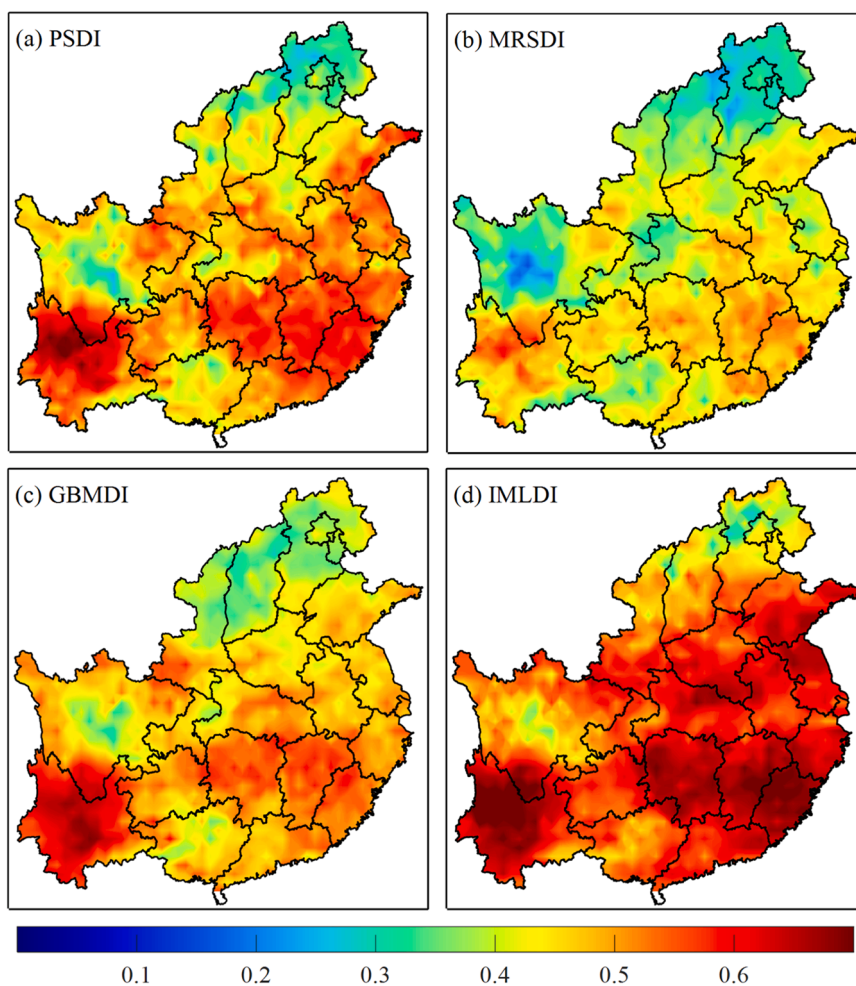


Fig. 6. Spatial distribution of correlation coefficients between PSDI, MRSDI, GBMDI, IMLDI and SPEI during 2001–2020.

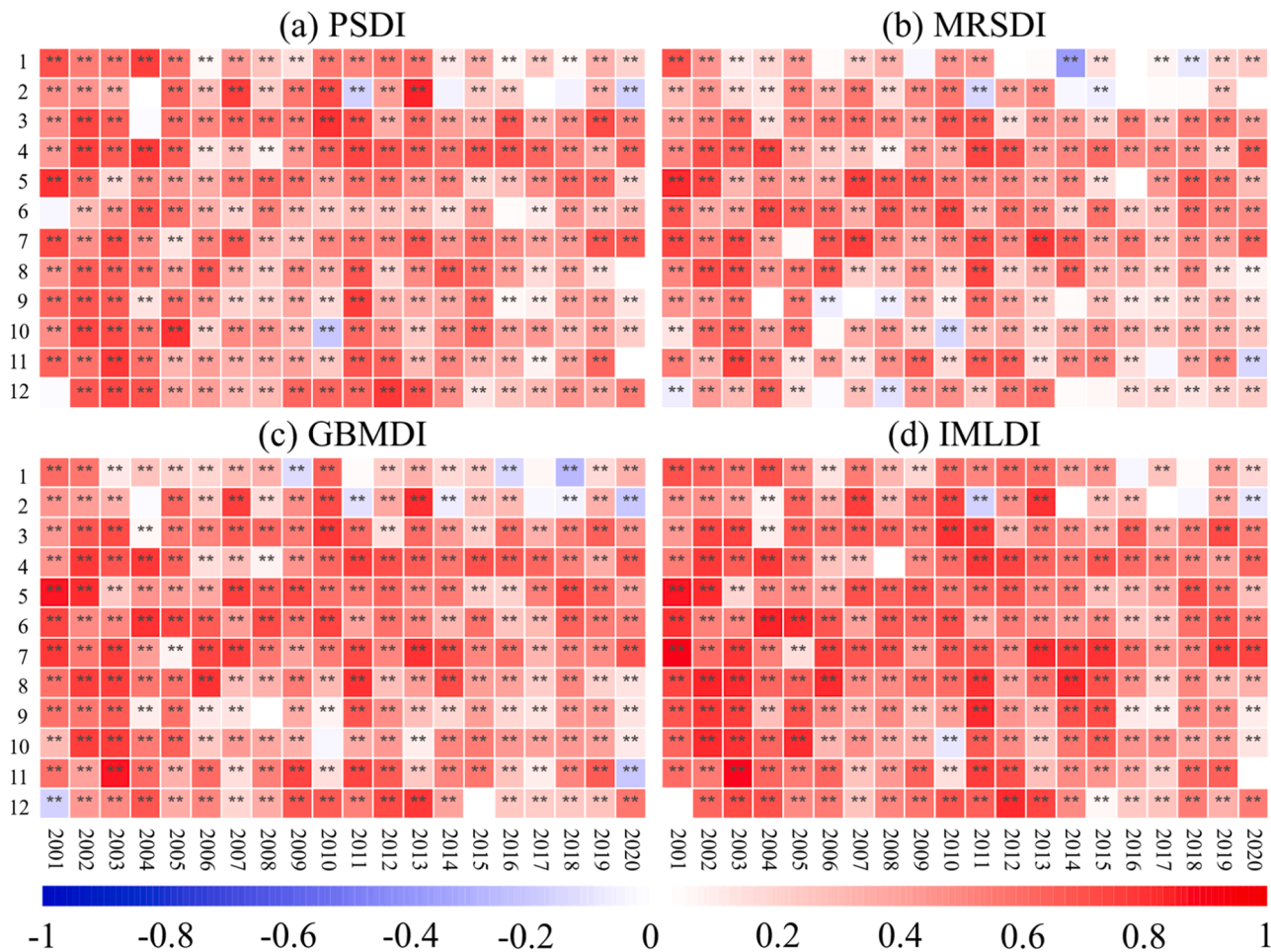


Fig. 7. Correlation coefficients for all months between PSDI, MRSDI, GBMDI, and IMLDI and SPEI from January 2001 to December 2020. ** denotes significant correlation at 0.01 significance level.

positive and negative contributions from each variable to drought in different land cover types (Figs. 4 and 5). A positive SHAP value suggests that the sample have a promote effect on SPEI, while a negative SHAP value means the opposite effect in the beeswarm plots. During April (Fig. 4), WBCI has the highest positive correlations with SPEI, followed by SMCI and TCI in forest, shrubland, sparse vegetation, rainfed cropland, and irrigated cropland. However, TCI is the most important contributors and has a positive contribution to SPEI in grassland. During August (Fig. 5), larger values of SMCI, WBCI, and TCI contribute to a bigger SPEI value in most land cover types, but TCI has a negative contribution to SPEI in shrubland and irrigated cropland.

3.2. Validation of interpretable machine learning drought index (IMLDI)

3.2.1. Comparisons of integrated remote sensing drought indices and SPEI

To assess the drought monitoring performance of IMLDI, we used the Pearson correlation analysis (Cohen et al., 2009) to compare correlation coefficients between integrated remote sensing drought indices (PSDI, MRSDI, GBMDI, and IMLDI) and SPEI. The spatial distributions of correlation coefficients between each integrated remote sensing drought index and SPEI from 2001 to 2020 are shown in Fig. 6. Generally, PSDI, MRSDI, GBMDI and IMLDI are positively correlated with SPEI, and the correlation coefficients are significant at 0.01 level in almost the eastern parts of China. These results indicate that PSDI, MRSDI, GBMDI, and IMLDI can well capture the observed drought. However, IMLDI has higher correlation coefficients (higher than the 0.4) with SPEI than PSDI, MRSDI, and GBMDI in most parts, especially in Southeast China

and Yunnan province. The spatial correlation coefficients between PSDI and MRSDI and SPEI are relatively low in most parts. This may be due to the fact that the PCA method omits some important information in the development of drought index (Greenacre et al., 2022; Hao et al., 2015; Kim et al., 2021), as well as the multiple linear regression model is based on the assumption of linearity and thus the model may not be fitted well when there are nonlinear relationships in input variables (Abdul Gafoor et al., 2022; Kim et al., 2020; Uyanık and Güler, 2013).

Moreover, we analyze the monthly spatial correlations between PSDI, MRSDI, GBMDI, and IMLDI and SPEI from 2001 to 2020 (Fig. 7). The percentages of months with correlation coefficients higher than 0.4 (average correlation coefficients) between PSDI, MRSDI, GBMDI, and IMLDI and SPEI are 58.33 % (0.433), 50.00 % (0.372), 60.00 % (0.439), and 71.67 % (0.508), respectively, from January 2001 to December 2020. The correlation coefficients are also significant at 99 % confident level in almost all months. The correlations analysis results indicate that IMLDI can better capture drought conditions than PSDI, MRSDI, and GBMDI.

3.2.2. Cross-validation with statistical drought-affected cropland areas and GPP

The annual drought index-based drought-affected cropland areas were counted in each province, were further compared to the real-world observed drought-affected cropland areas. For each integrated remote sensing drought index, the grids of cropland in which moderate, severe, or extreme droughts occurs for more than two consecutive months in a year were identified as the annual drought index-based drought-affected

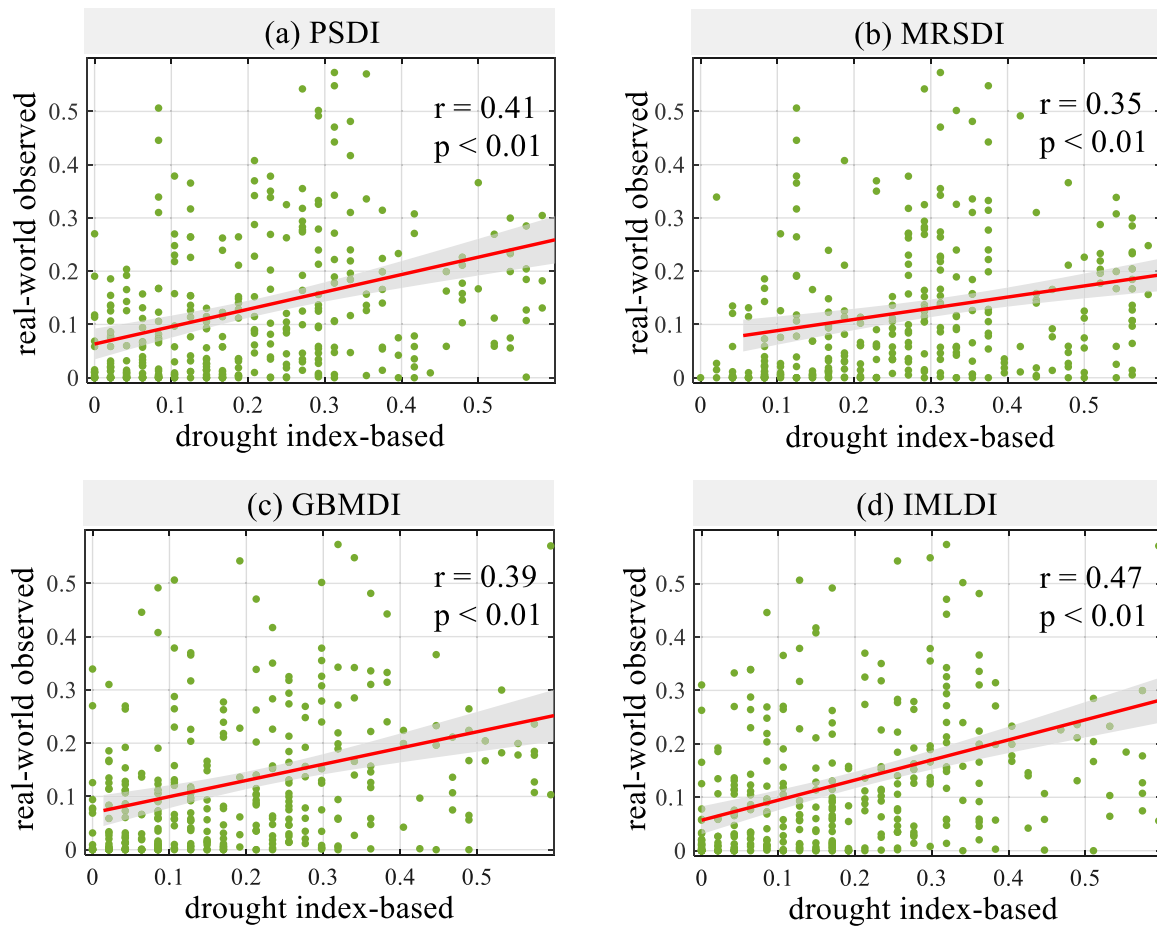


Fig. 8. The scatter plots and correlations of annual real-world observed and drought index-based drought-affected cropland areas. The drought-affected cropland areas are scaled.

cropland areas (Rivera and Penalba, 2014; Sun et al., 2017). It can be seen from Fig. 8 that the correlations between the scaled drought-affected cropland areas monitored by PSDI, MRSDI, GBMDI, and IMLDI and the scaled real-world observed drought-affected cropland areas are higher than 0.35, and significant at 99 % confident level. Compared to PSDI (0.41), MRSDI (0.35) and GBMDI (0.39) indices, the drought-affected cropland areas monitored by IMLDI (0.47) have a higher correlation coefficient with the real-world observed drought-affected cropland areas.

The spatial correlations of integrated remote sensing drought indices and GPP are also compared (Fig. 9). PSDI and MRSDI have a positive correlation with GPP ($p < 0.01$) in Guangxi province, Guizhou province, and the northern parts. GBMDI shows a positive correlation with GPP ($p < 0.01$) only in Guangxi and Guizhou provinces. Compared with PSDI, MRSDI, and GBMDI, IMLDI is significantly positive correlated with GPP in much larger areas, which covers almost the entire study area. In terms of each province, GPP is significantly correlated with IMLDI in most provinces. The number of provinces with significant correlation (the value of correlation coefficients) of IMLDI and GPP is larger (higher) than that of PSDI, MRSDI, and GBMDI and GPP. To further validate the capabilities of reflecting GPP of crops, we analyze the spatial correlations of integrated remote sensing drought indices and GPP with a special focus on agricultural land (Supplementary Fig. S1). We found that the correlations of IMLDI and GPP are significant in almost all agricultural land, and the correlation coefficients are also higher than other three drought indices in most agricultural land.

3.2.3. Two typical drought events monitoring using integrated remote sensing drought indices

Southwestern China suffered a severe drought from autumn 2009 to spring 2010. According to the survey of the Ministry of Civil Affairs, around 21 million people were short of drinking water, and economic losses reached nearly US\$30-billion during this drought event (Lu et al., 2011; Yang et al., 2012; Zhang et al., 2012). In 2019, drought disasters occurred in 24 provinces in China, and the area of crops affected by drought reached 8777.83 thousand hectares (Ding and Gao, 2020; Yan et al., 2021). Particularly, serious drought occurred in Yunnan province and Huang-Huai-Hai Region during April to August 2019, which have caused great damage to local vegetation. Therefore, these two severe drought events were selected to further evaluate the performance of IMLDI in monitoring drought evolutions.

We analyzed the spatial distributions of PSDI, MRSDI, GBMDI, and IMLDI during August 2009 to June 2010 (Figs. 10 and 11), and validated them by SPEI. Drought conditions indicated by SPEI were initially located in much eastern parts of China during August 2009 and drought center appeared in Southwest China in September 2009. After that, drought became more severity and affected more areas in Southwest China during October 2009, continued to April 2010, and ended in June 2010. The spatial distributions of PSDI, MRSDI, GBMDI, and IMLDI generally agree with SPEI in most parts during the stages of onset, progression, and end of 2009/2010 drought. However, MRSDI and GBMDI overestimate the spatial extent and severity of drought indicated by SPEI in August 2009 and October 2009. Drought affected areas observed by PSDI are relatively fragmented, especially in August 2009 and June 2010, which is inconsistent with regional spatial aggregation of drought. Moreover, PSDI, MRSDI, and GBMDI underestimate the

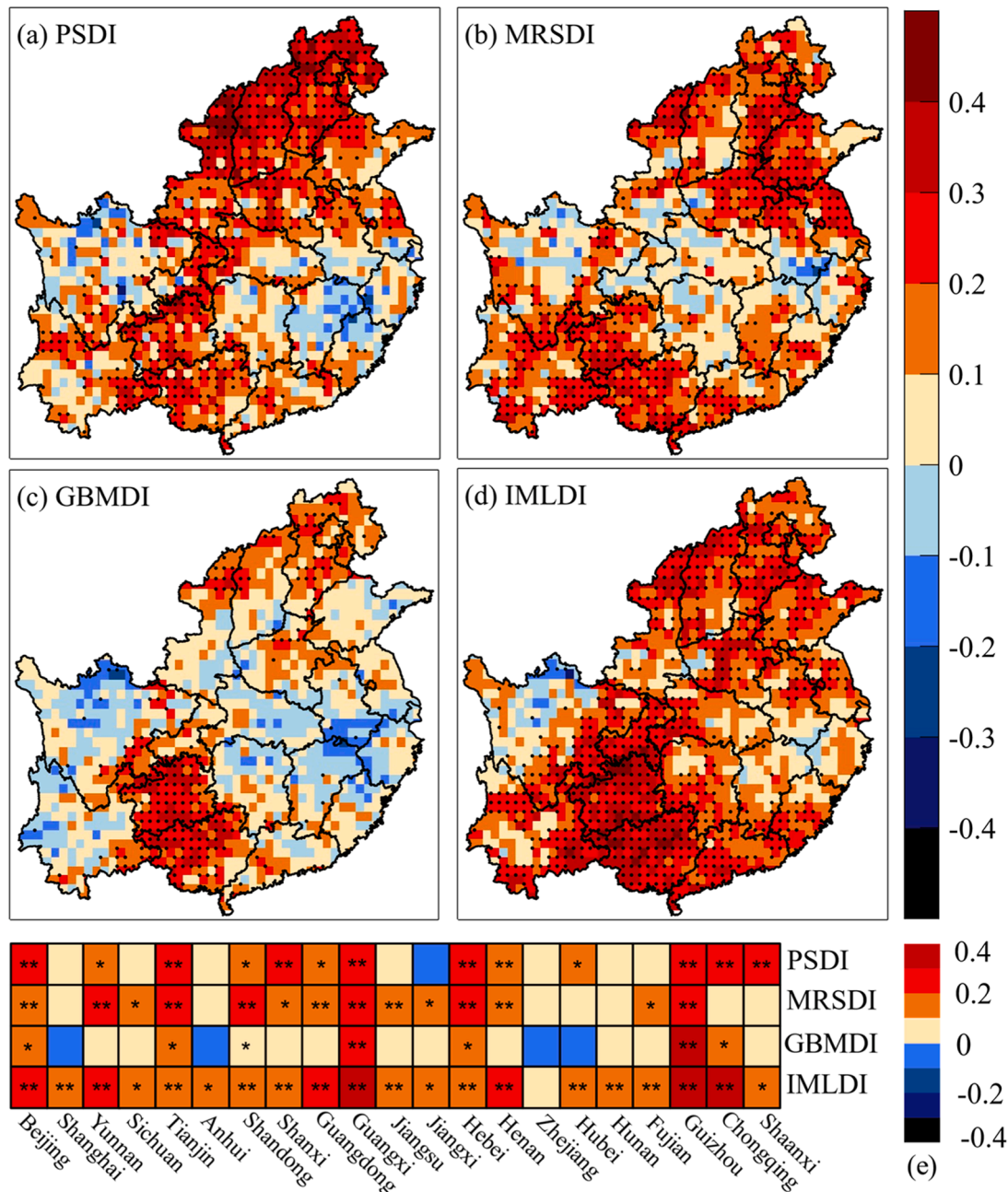


Fig. 9. Spatial distributions of correlation coefficients between PSDI, MRSDI, GBMDI, IMLDI and GPP. The stippling areas indicate that the correlations are significant at the 99 % confidence level. * and ** denote that the correlations are significant at 95 % and 99 % confidence levels, respectively.

drought affected areas in Southwest China in April 2010. Overall, compared with PSDI, MRSDI, and GBMDI, IMLDI can better capture the evolutions of 2009/2010 drought.

In 2019 (Fig. 12), drought centers indicated by SPEI are located in Yunnan province and Huang-Huai-Hai Region. PSDI, MRSDI, GBMDI and IMLDI can capture the onset, progression, and end of this drought event to a certain extent. However, MRSDI and GBMDI cannot capture drought conditions in July in Yunnan province. MRSDI (PSDI) overestimated (underestimated) the severity and affected areas of drought in Huang-Huai-Hai Region during May to July 2019, which may be explained by the fact that MRSDI or PSDI failed to fully integrate all the information from the input remote sensing variables (Zhang et al., 2022). The integrated remote sensing drought indices generally underestimate the severity of drought because agricultural drought are always less severe than meteorological drought in the same area during the

same time periods (Degefu, 1987; Hao et al., 2015). In summary, drought conditions monitored by IMLDI are more consistent with the actual situations, and thus IMLDI is more suitable for regional drought monitoring.

3.3. IMLDI-based agricultural drought risk analysis in the eastern parts of China

Spatial distributions of frequency, average duration, average severity, and average peak of agricultural drought during 2001–2020 are given in Fig. 13. Agricultural drought tends to occur frequently in East China, but the values of duration, severity, and peak of drought are relatively low. More than 25 agricultural drought events are detected in Huang-Hai Region and Yunnan province, and the duration, severity, and peak of drought are quite high. The values of frequency, duration,

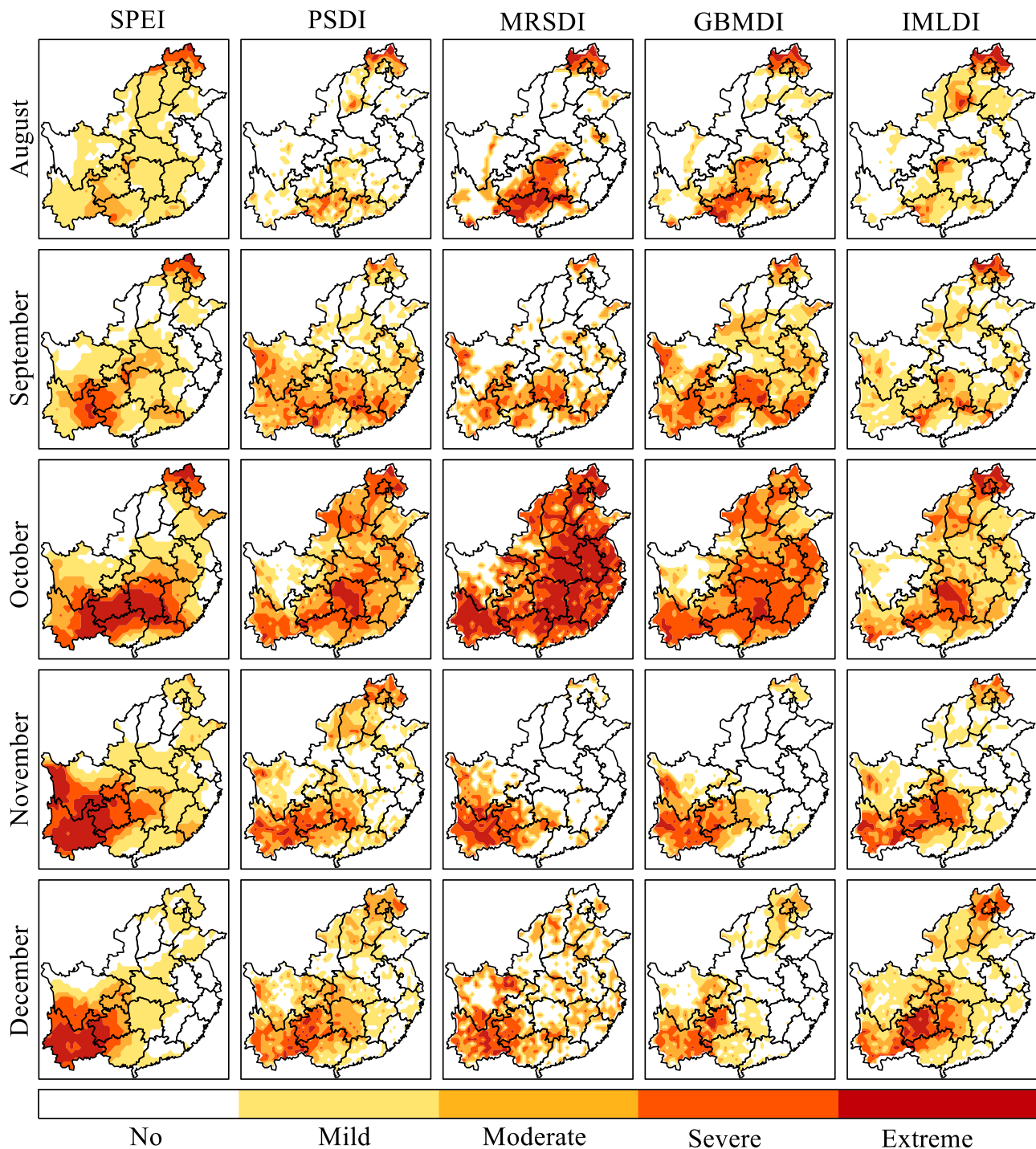


Fig. 10. Spatial and temporal evolutions of drought monitored by SPEI, PSDI, MRSDI, GBMDI, and IMLDI during August 2009 to December 2009.

severity, and peak of drought are high in Guizhou and Guangxi Provinces. These results indicate that Huang-Hai Region and Yunnan, Guizhou, and Guangxi Provinces suffered from severe agricultural droughts frequently, which is consistent with the results of Wu et al. (2022).

To analyze the occurrence risks of more severe agricultural droughts, we analyzed the univariate and bivariate return periods for duration ($D > 14$), severity ($S > 2.2$) and peak ($P > 0.45$) of drought events in the eastern parts of China (Fig. 14). The marginal distributions of the duration, severity, and peak of agricultural drought were fitted by the exponential, gamma, and Pareto cumulative distribution functions, respectively. Bivariate joint distribution functions were selected from the best-fitting Archimedean copula functions of Clayton, Frank, and

Gumbel. The results show that a relatively short return period (approximately 5–15 years) of drought with duration more than 14 months or severity higher than 2.2 is mainly detected in Huang-Hai Region and Yunnan, Guizhou, and Guangxi Provinces (Fig. 14a and b). The short return period of peak ($P > 0.45$) is mainly located in Guangdong, Guangxi, and Guizhou provinces (Fig. 14c). Moreover, the logical 'or' (Fig. 14d-f) and 'and' (Fig. 14g-i) bivariate return period analysis also show that the short return periods of duration ($D > 14$), severity ($S > 2.2$) and/or peak ($P > 0.45$) of drought events are observed in Huang-Hai Region, and Yunnan, Guizhou, and Guangxi Provinces. Overall, the return period analysis suggests that Huang-Hai Region and Yunnan, Guizhou, and Guangxi Provinces have a high risk to suffer from severe

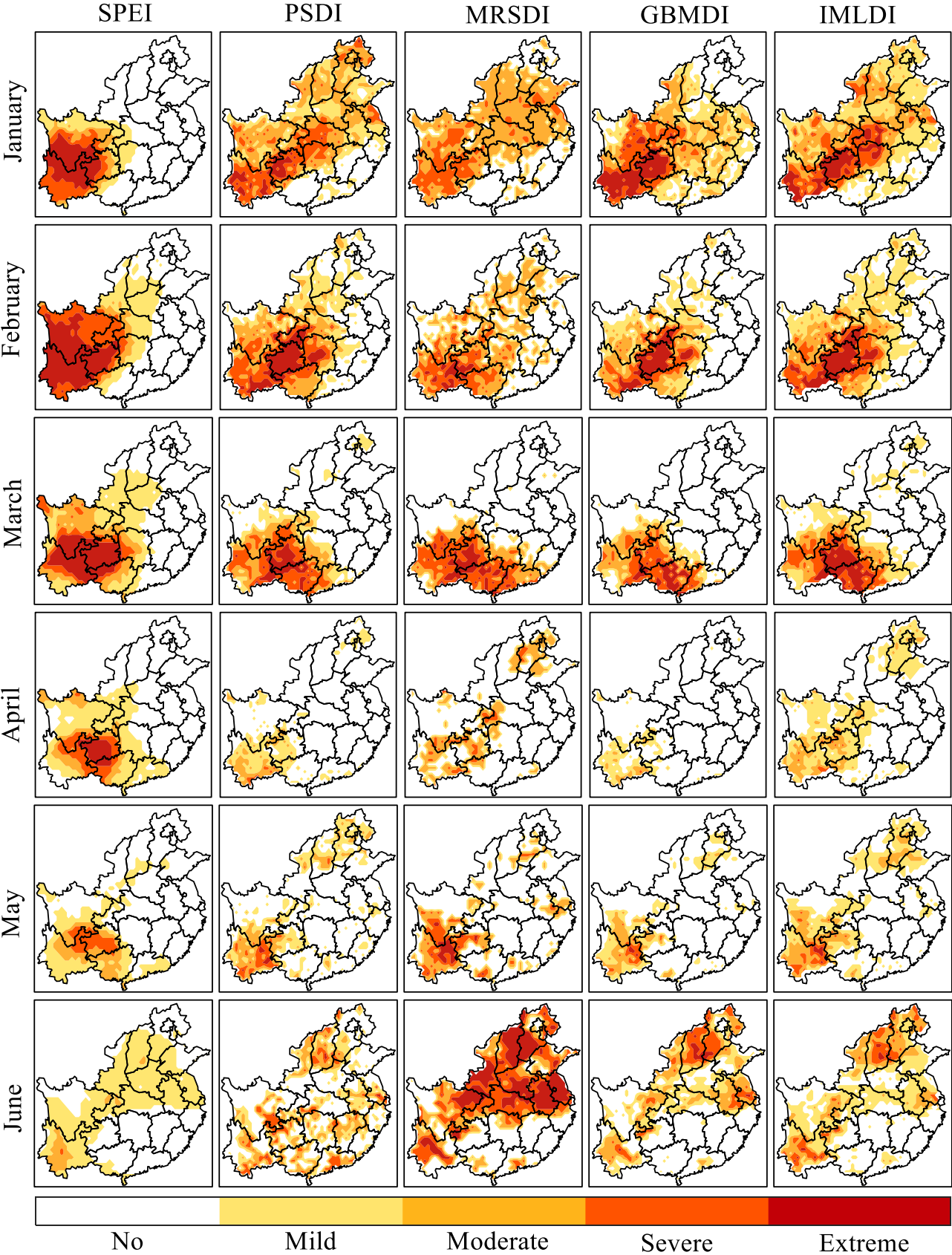


Fig. 11. Spatial and temporal evolutions of drought monitored by SPEI, PSDI, MRSDI, GBMDI, and IMLDI during January 2010 to June 2010.

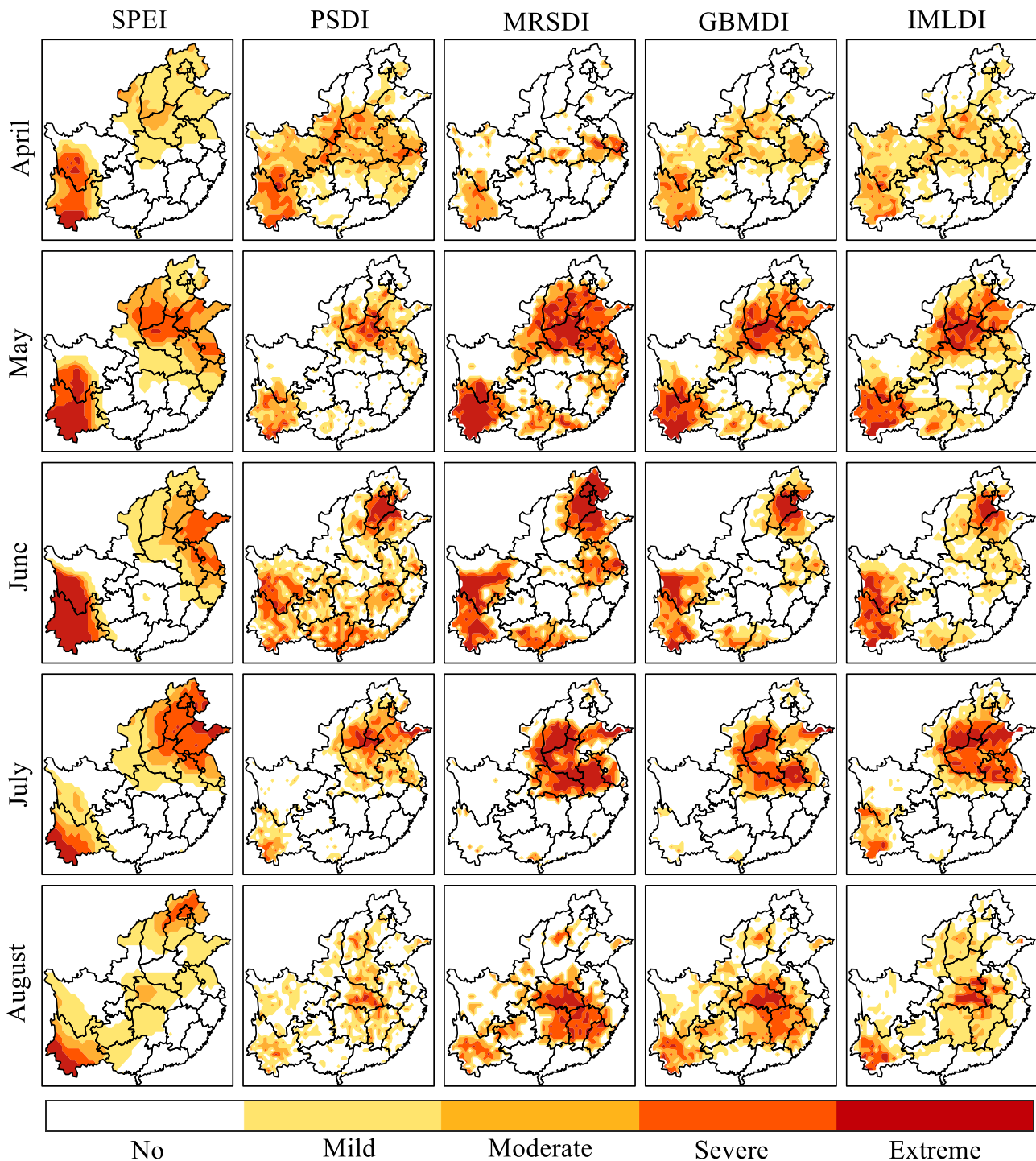


Fig. 12. Spatial and temporal evolutions of drought monitored by SPEI, PSDI, MRSDI, GBMDI, and IMLDI during December April 2019 to August 2019.

agricultural droughts.

4. Discussion

During recent decades, droughts become more frequent and intensified due to climate warming (Dai et al., 2020; Niu et al., 2023), which poses a serious threat to the agriculture, ecosystem, and sustainable infrastructure and resource management (Hussein and Basel, 2024; Valjarević et al., 2022). Drought index is a way to quantitatively describe drought characteristics and facilitate drought risk analysis and monitoring. NDVI, P, SM, and LST retrieved from remote sensing are

most widely used to develop integrated drought indices, such as GIDI and CDI. However, previous studies suggested that the response of NDVI to drought stress has a considerably long lag time (Song et al., 2018; Sun et al., 2015), and P cannot accurately reflect climatic wet and dry conditions. As SIF can reflect the physiological characteristics of vegetation and WB affects vegetation absorbing water from the soil, NDVI and P were replaced by SIF and WB, respectively, to develop IMLDI in this work, which may improve the performance of agricultural drought monitoring (Figs. 6–12). Previous study also used SIF, WB, and LST to develop TSWDI, and found it can better describe drought conditions than TVPDI based on NDVI, P, and LST (Liu et al., 2023b).

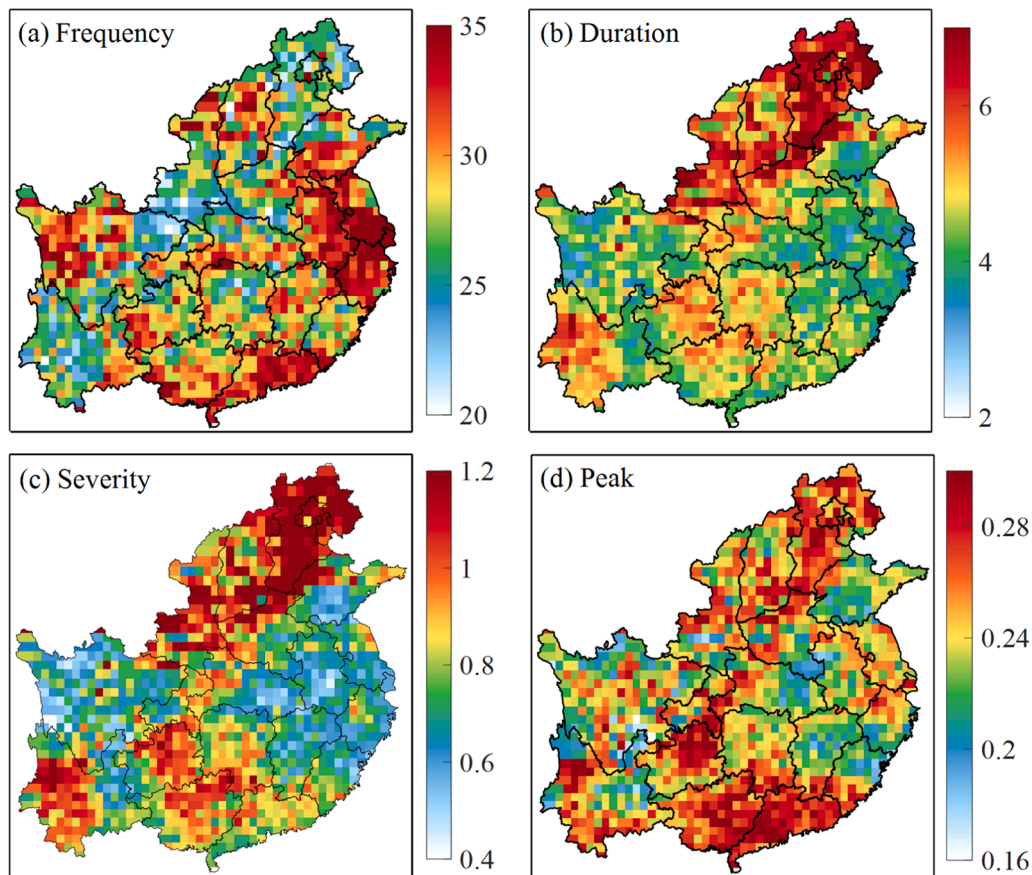


Fig. 13. Frequency (a), average duration (b, month), average severity (c), and average peak (d) of agricultural drought during 2001–2020.

The weights of each variable in drought index are the key to accurately monitoring drought (Guo et al., 2019). Previous studies used simple linear combinations to construct integrated drought indices, such as TVPDI, TSWDI, MIDI, and Optimal Scaled Drought Condition Index (OSDCI) (Guo et al., 2019). Simple linear combination models have the advantage of being relatively easy to compute and simple to implement (Hao and Singh, 2015; Jiao et al., 2019), but the weights of input variables are determined subjectively. In recent years, several studies have used machine learning methods to calculate the weights of input variables (Jiao et al., 2021; Xu et al., 2023), which is more suitable for describing complex drought phenomena. The weights of Composite Drought Monitoring Index (CDMI) and Comprehensive remote sensing-based Agriculture Drought Condition Indicator (CADCI) were determined by the random forest method (Alkaraki and Hazaymeh, 2023; Hanadé Houmma et al., 2022), and the GBMDI was constructed based on the gradient boosting method (Zhang et al., 2022). Compared with other widely used drought indices, these integrated drought indices have a better performance in drought monitoring. However, there are still issues with hyperparameter optimization, overfitting, and black-box characteristics. Our work combines Bayesian-optimized LGBM and SHAP to obtain accurate and interpretable results for the development of IMLDI. The results (Figs. 6–12) from drought indices comparisons, cross-validation, and typical drought events monitoring indicate that IMLDI is a more reliable drought index for agricultural drought monitoring.

The application of IMLDI can be extended in other regions based on different remote sensing datasets and land cover types. If climates and land cover types in other regions are quite different from the study area of this work, the weights of input remote sensing variables should be recalculated and the performance of IMLDI should be further validated. All remote sensing datasets used in the development of IMLDI are public

available in a near-real time. Moreover, IMLDI was developed based on R and RStudio, which are also open source and freely available software, and can be easily reproduced by practitioners with a certain professional knowledge. Therefore, IMLDI can be used to monitor agricultural drought in a near-real time.

Despite all these advantages, there are some limitations that may affect the performance of IMLDI. Firstly, the development of IMLDI relies heavily on multi-source remote sensing data. However, remote sensing is susceptible to atmospheric water vapors, aerosols, cloud cover, and sensor degradation, and usually has data gaps in some regions (Rousta et al., 2021). Therefore, remote sensing data has certain observation uncertainties, which may directly lead to accumulated uncertainties in drought monitoring of IMLDI. Secondly, as the correlations of annual real-world observed and drought index-based drought-affected cropland areas may be affected by various factors, such as the quality of statistical areas and spatial resolution of drought index, the results of validation based on annual statistical drought-affected cropland areas should be interpreted with caution. In future research, the performance of IMLDI should be tested against a broader range of real-world scenarios or field-based drought indices (e.g., SPI, PDSI), which may further verify the reliability of IMLDI in drought monitoring.

5. Conclusion

Due to the complexity of drought disasters, a single or two variables-based drought indices cannot fully reflect the real situation of agricultural drought. The development of integrated multiple variables drought indices is still urgent needed, and can benefit from the application of machine learning. In this work, a new agricultural drought index (IMLDI) was developed based on multi-source remote sensing data (SIF, WB, SM, and LST) and interpretable machine learning. The performance

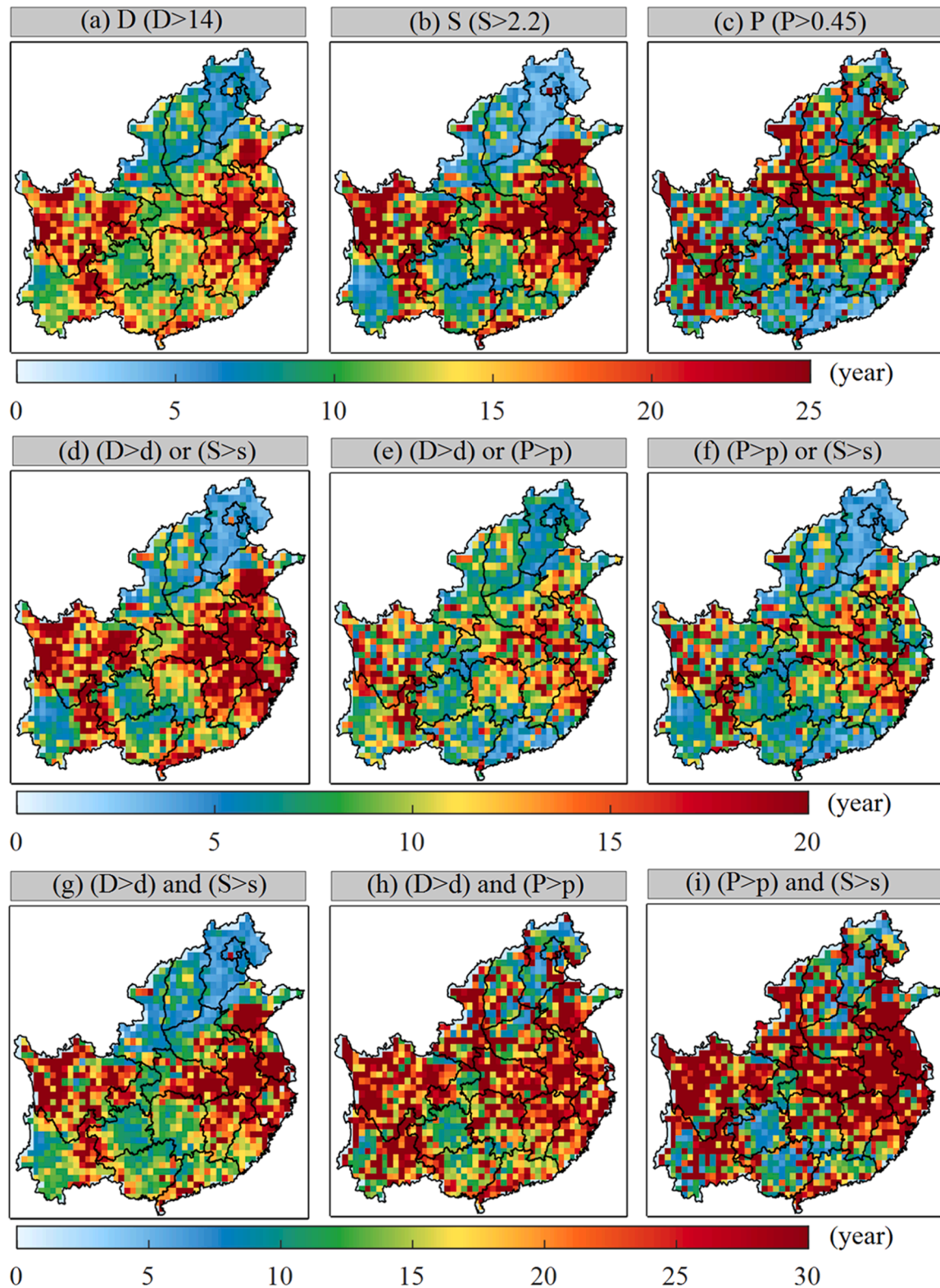


Fig. 14. Spatial distributions of univariate and bivariate return periods for drought duration ($D > 14$, month), severity ($S > 2.2$) and peak ($P > 0.45$). a, b, and c indicate univariate return periods of drought duration, severity, and peak. d, e, and f are bivariate logic 'or' joint return periods. g, h, and i are bivariate logic 'and' joint return periods.

of IMLDI was systematically evaluated using SPEI, three widely used drought indices, the real-world drought-affected cropland areas, GPP, and two severe drought events. Besides, IMLDI-based drought risks were also analyzed in the eastern parts of China during 2001–2020.

Compared with PSDI, MRSDI, and GBDMI, IMLDI can better capture drought conditions indicated by SPEI, has higher correlation coefficients with the real-world observed drought-affected cropland areas, and is significantly positive correlated with GPP in much larger areas (almost

in the entire regions). IMLDI can also successfully detect the drought dynamic changes of 2009/2010 and 2019 drought events, and thus IMLDI can be well employed to monitor agricultural drought. The Huang-Hai Region and Yunnan, Guizhou, and Guangxi Provinces experienced more than 25 severe agricultural drought events during 2001–2020, and the return period analysis also suggests that these regions have a high risk to suffer from severe agricultural droughts. In summary, this work provides a new and useful agricultural drought index based on multi-source remote sensing data, and has a great potential to improve agricultural drought monitoring in regions with limited ground-based observations.

CRedit authorship contribution statement

Hao Chen: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ni Yang:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition, Formal analysis. **Xuanhua Song:** Visualization, Validation, Methodology, Investigation, Data curation. **Chunhua Lu:** Writing – review & editing, Software, Methodology, Investigation. **Menglan Lu:** Writing – review & editing, Methodology, Data curation. **Tan Chen:** Writing – review & editing, Visualization, Conceptualization. **Shulin Deng:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.agwat.2025.109303.

Data availability

The data that has been used is confidential.

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